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Module - 2

Quantum Mechanics

Presentation

By

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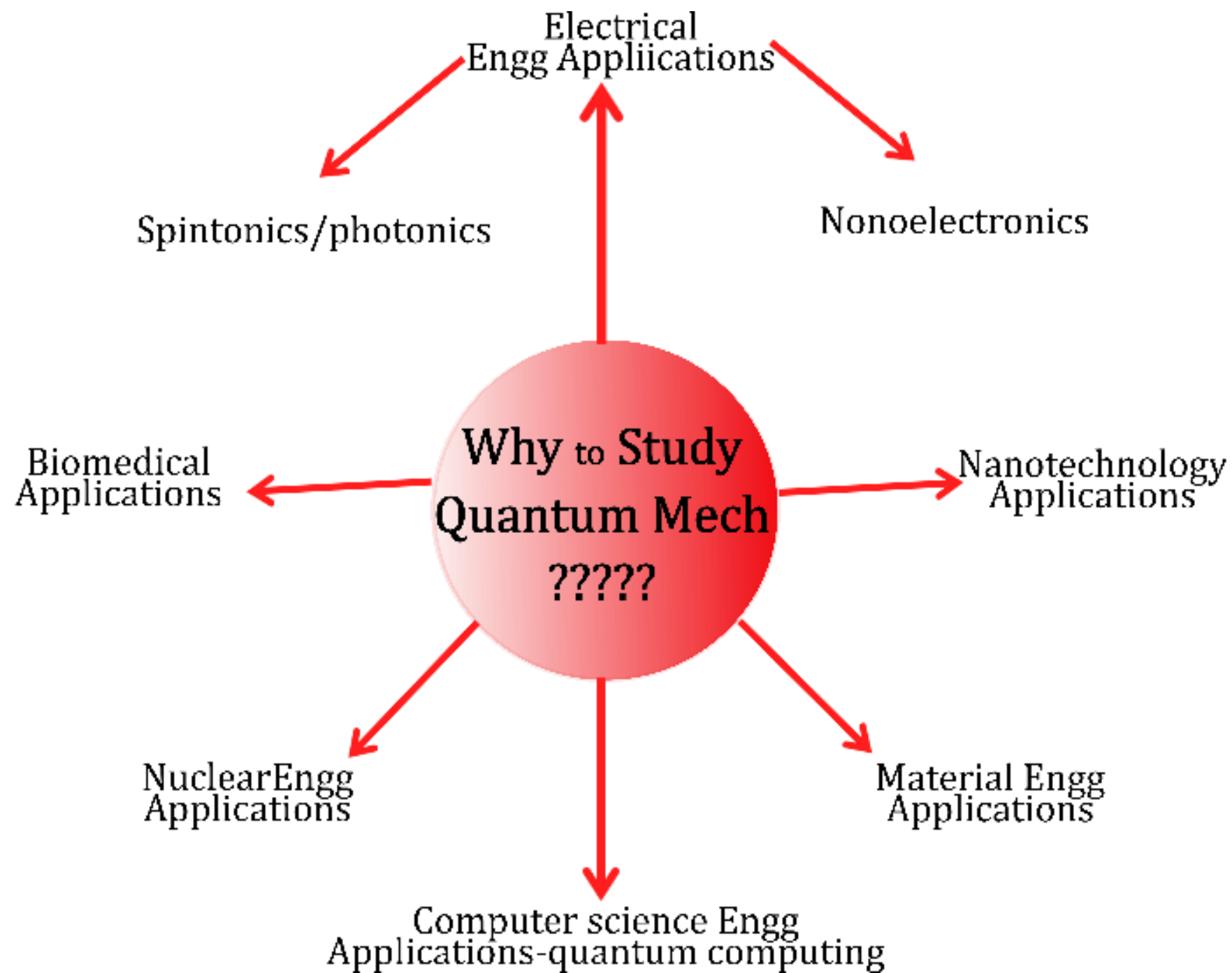
Department of Physics

Quantum Mechanics

de Broglie Hypothesis and Matter Waves, de Broglie wavelength and derivation of expression by analogy, Phase Velocity and Group Velocity, **Heisenberg's Uncertainty Principle** and its application (Non existence of electron inside the nucleus - Non Relativistic), Principle of Complementarity,

Wave Function, **Time independent Schrödinger** wave equation (Derivation), Physical Significance of a wave function and Born Interpretation, Expectation value, Eigen functions and Eigen Values, **Particle inside one dimensional** infinite potential well, Quantization of Energy States, Waveforms and Probabilities.

Numerical problems- de Broglie Hypothesis, Heisenberg's Uncertainty Principle, Energy Eigen Values for a particle in 1D infinite potential well



History: *At the end of 19th century scientists felt that all the laws of physics (which were known at that period of time) are enough to explain all the events occur in nature. It was believed that there are only two kinds in nature. First the bodies which are made up of particles and second radiation. All particles obey Newton's laws of motion and radiation obey Maxwell's equations of electromagnetism. These laws are now a days known as classical physics.*

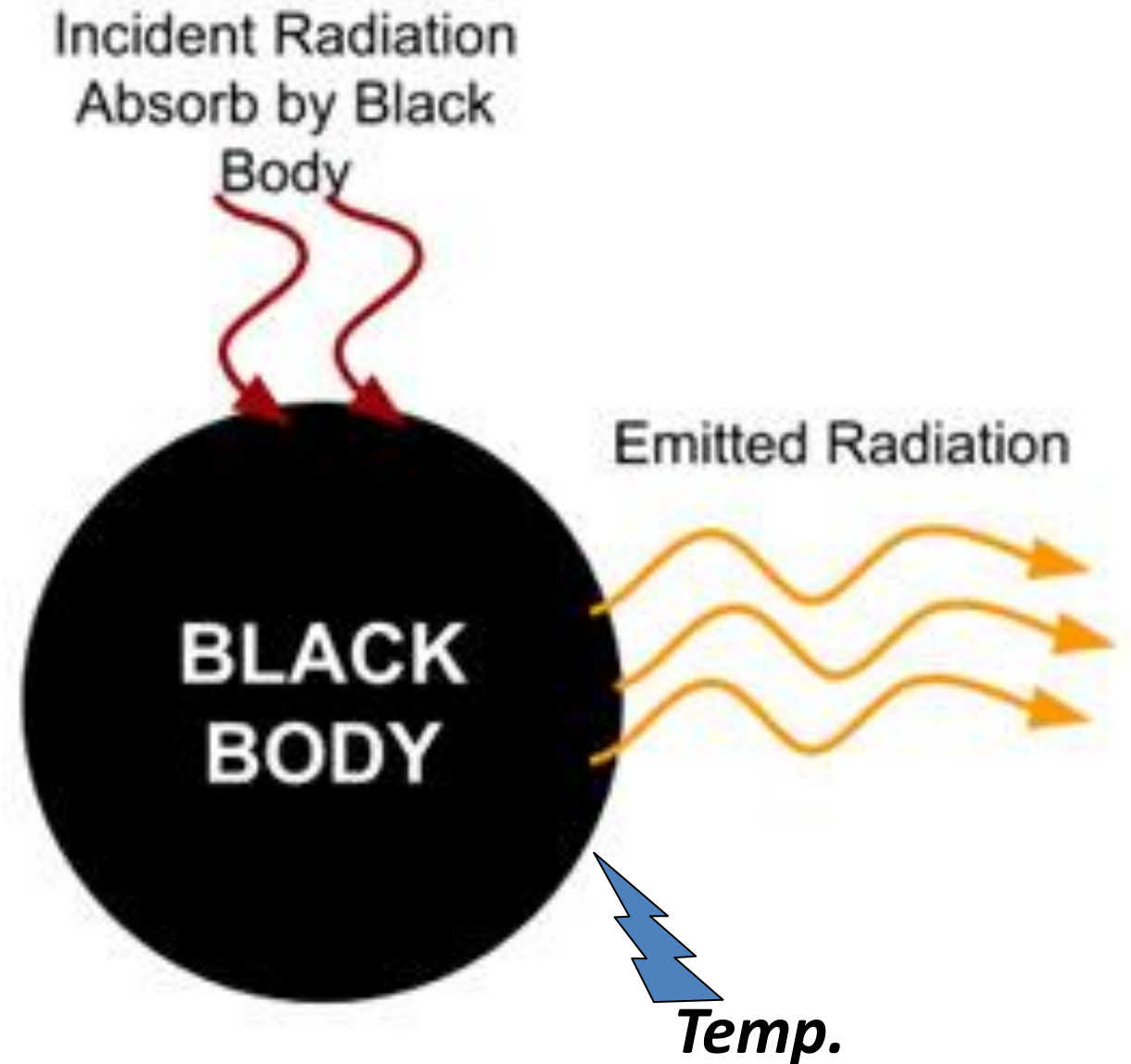
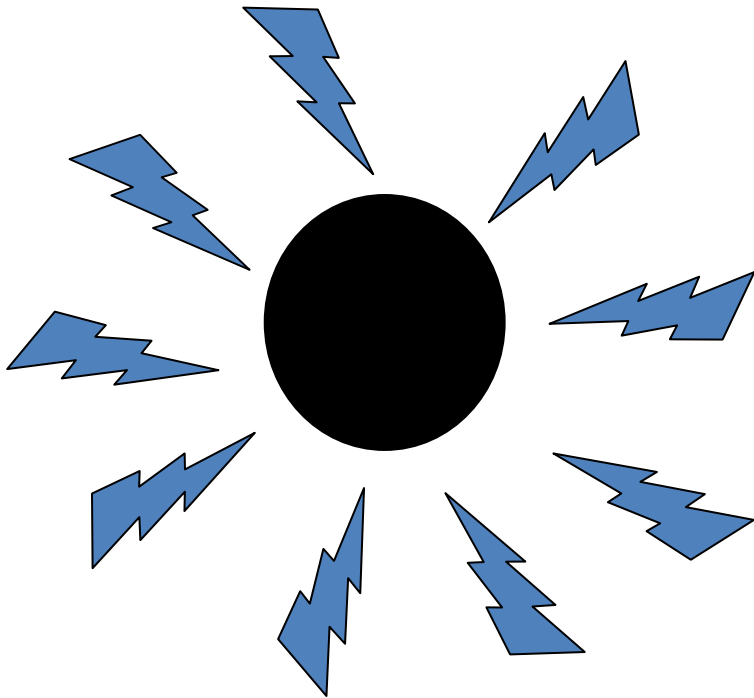
Fortunately at the same time some experiments were performed and the experimental results were not in comparison with so called classical theory. Some of them are Black body radiation, Photo electric effect and Compton Effect etc. In order to overcome the drawbacks of Classical Physics a new types of laws of physics have been introduced which are now known as Quantum Physics. Black body radiation is one of the great effects which has led to the concept of Modern Physics-quantum physics.

✓ Why Quantum Mechanics need?

- By the end of the 19th century it seemed that all the laws of physics were known.
- the motion of the planets was understood (Macro Size).
- However, there were a few problems where classical physics (pre-20th century) fails to explain (Micro).
 - ✓ Atomic level
 - ✓ Blackbody radiation
 - ✓ Photoelectric effect
 - ✓ light was particles or Wave ?

✓ What do u mean by Quantum mechanics?

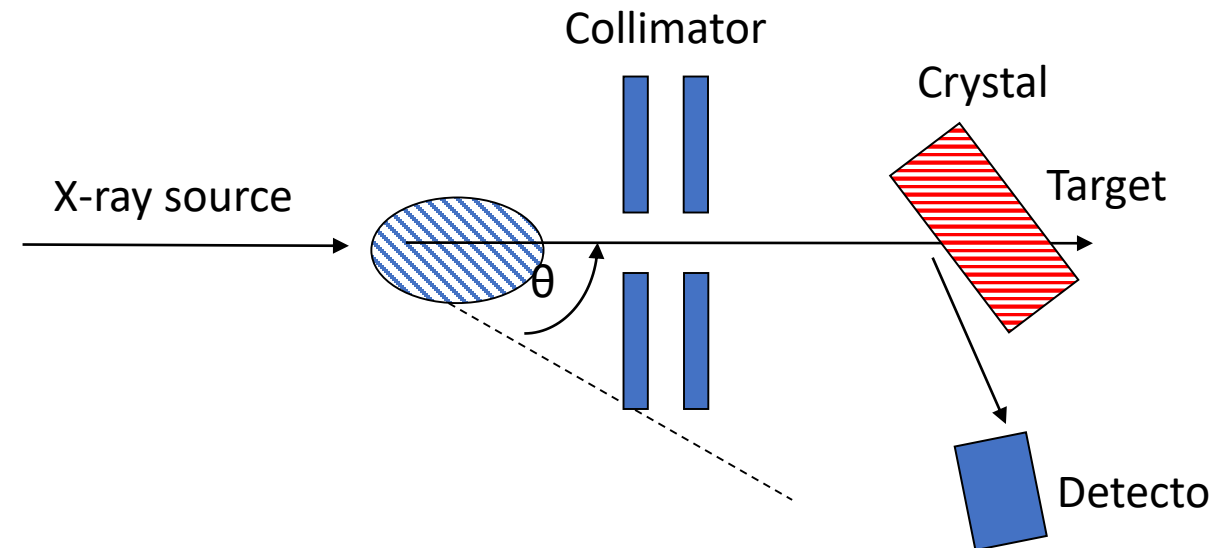
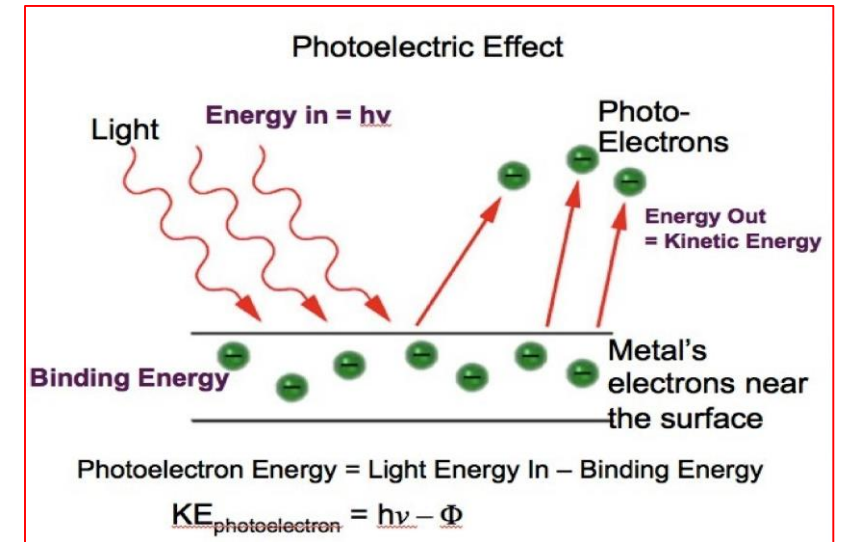
Black Body:



Wave-Particle dualism

Evidence for wave-particle duality

- Photoelectric effect
- Compton effect
- Electron diffraction
- Interference of matter-waves



Consequence: de Brooglie Hypothesis & Heisenberg uncertainty principle

Wave-Particle dualism- de Broglie Hypothesis

- ✓ The origin of quantization of energy lies in the dual nature.
- ✓ For example, **Compton** observed that X-rays are scattered like **particle** by an electron and **Brag** observed that X-rays get diffracted by a crystal like a **wave**.

de Broglie Hypothesis (1924): “Just like radiation, that behaves as a particle as well as wave under different circumstances, then entities (say electrons in matter) which ordinarily behaves as particle can exhibit wave properties under appropriate circumstances”.

$$i. e., \quad \lambda = \frac{h}{p}$$



Matter waves : A wave is associated with each moving particle which is called a matter-wave.

Experimental proof-Wave nature of electron:- Davisson & Germer of USA (in 1927) accidentally conducted an experiment to show diffraction of accelerated electrons from nickel crystal just like the way. This diffraction of electrons confirms their wave nature.



Derivation of expression for de Broglie Wavelength (λ)

Consider a photon of mass ' m ' moving with velocity ' c ', then the energy of photon is given by,

$$E = mc^2 \quad (1)$$

According to Plank's theory of radiation, the energy of photon with frequency γ is given by

$$E = h\gamma \quad (2)$$

Equating **eqn, (1)** and **(2)**, we have,

$$mc^2 = h\gamma \quad \Rightarrow \quad mc^2 = h \frac{c}{\lambda} \quad \because \quad \gamma = \frac{c}{\lambda}$$

Or,

$$\lambda = \frac{h}{mc}$$

But, $p = mc$,

$$\lambda = \frac{h}{p}$$

This is the expression for de-Brooglie wavelength

Different forms of de Broglie Wavelength (λ)

1)

$$\lambda = \frac{h}{\sqrt{2meV}}$$



Proof: Wkt, energy of an electron accelerated under a potential difference V is ' eV '. If ' m ' is the mass and ' v ' is the velocity of electron then the energy equation is given by,

$$eV = \frac{1}{2}mv^2 \quad (1)$$

If P is the momentum of the electron, then, $P = mv$

$$P^2 = m^2v^2 \Rightarrow mv^2 = \frac{P^2}{m} \quad (2)$$

Substituting **eqn, (2)** in **(1)**, we have,

$$eV = \frac{P^2}{2m} \Rightarrow P = \sqrt{2meV} \quad (3)$$

From de Broglie hypothesis, wkt, $\lambda = \frac{h}{p}$

$\Rightarrow$

$$\lambda = \frac{h}{\sqrt{2meV}}$$

$$\text{or } \lambda = \frac{1.226}{\sqrt{V}} \text{ nm}$$

2)

$$\lambda = \frac{h}{\sqrt{2mqV}}$$

3)

$$\lambda = \frac{h}{\sqrt{2mE}}$$

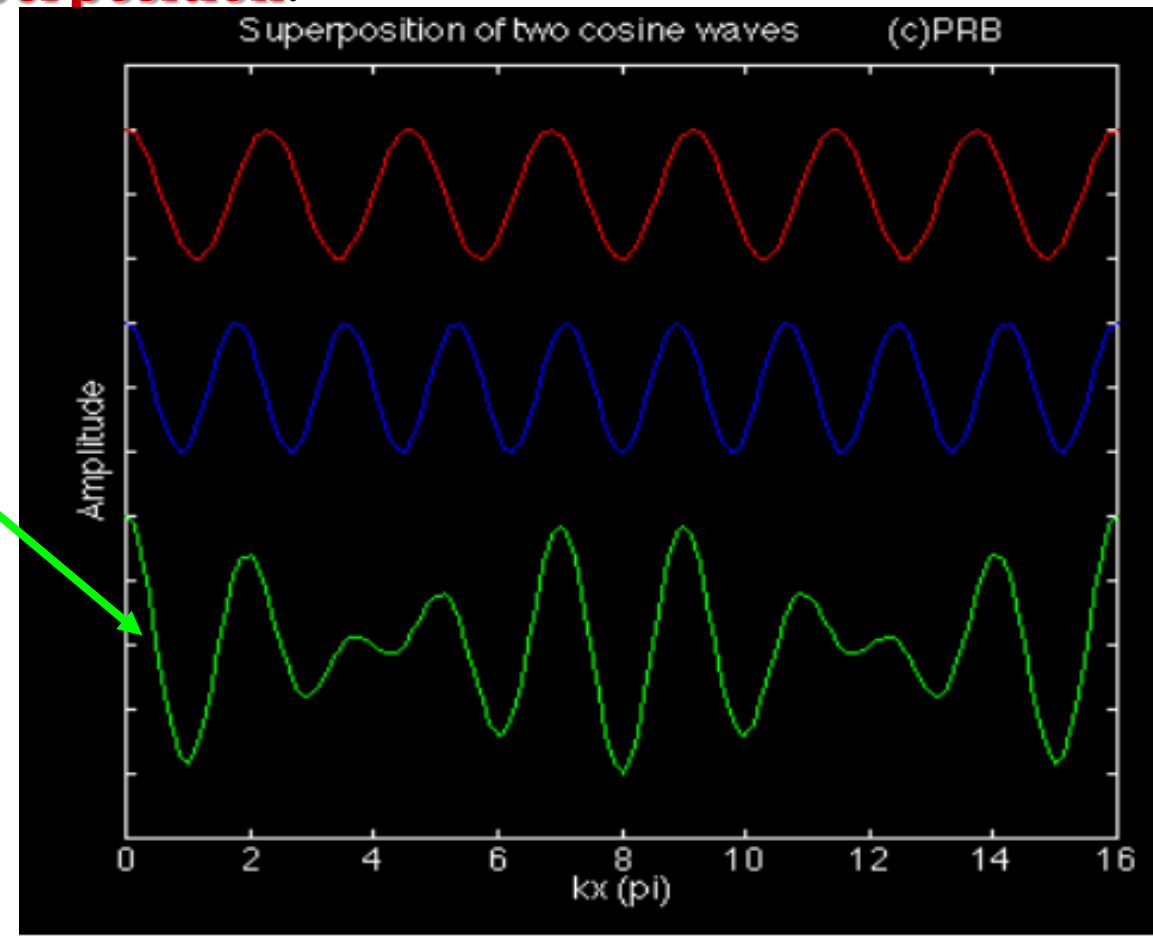
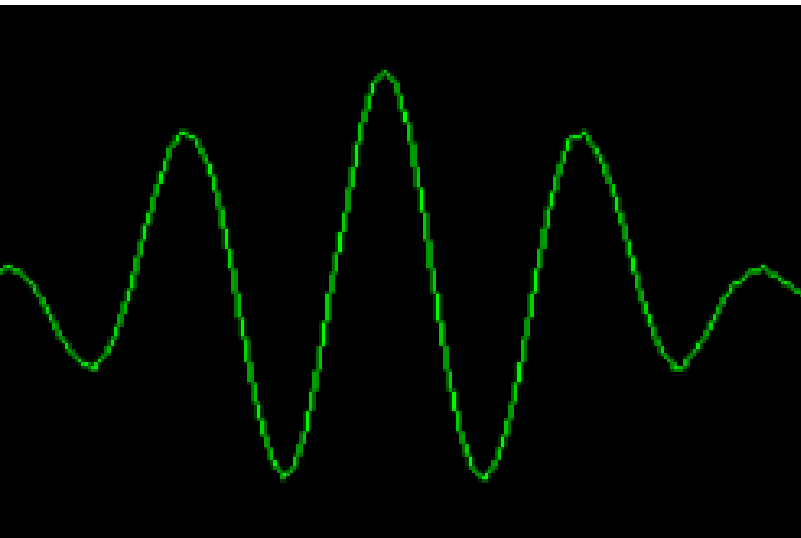
Wave Packets

Defⁿ: A **wave packet** refers to the case where two (or more) waves exist simultaneously.

A wave packet is often referred to as a wave group.

- This situation is permitted by the **principle of superposition**.

The green curve shows the sum of the two waves. It is the **wave packet**.



Wave Velocity (Phase Velocity) and Group Velocity

- Waves can be in a group and such groups are called **wave packets**,

Wave (Phase) velocity (V_{ph}) : The velocity with which individual wave travels in group of waves (Wave packet) is called **phase velocity**. or

In a travelling wave, the velocity with which the phase of wave is propagated is called **phase velocity**.

$$V_{Ph} = \frac{\omega}{k}$$



The phase velocity is equal to; $V_{ph} = \gamma \lambda$ -----(1)

But, wkt. angular frequency ; $\omega = 2\pi\gamma$ or $\gamma = \frac{\omega}{2\pi}$ -----(2)

& wave propagation vector is; $k = \frac{2\pi}{\lambda}$ or $\lambda = \frac{2\pi}{k}$ -----(3)

Substituting eqn (2)and (3) in (1), we get

$$V_{Ph} = \frac{\omega}{2\pi} \times \frac{2\pi}{k}$$



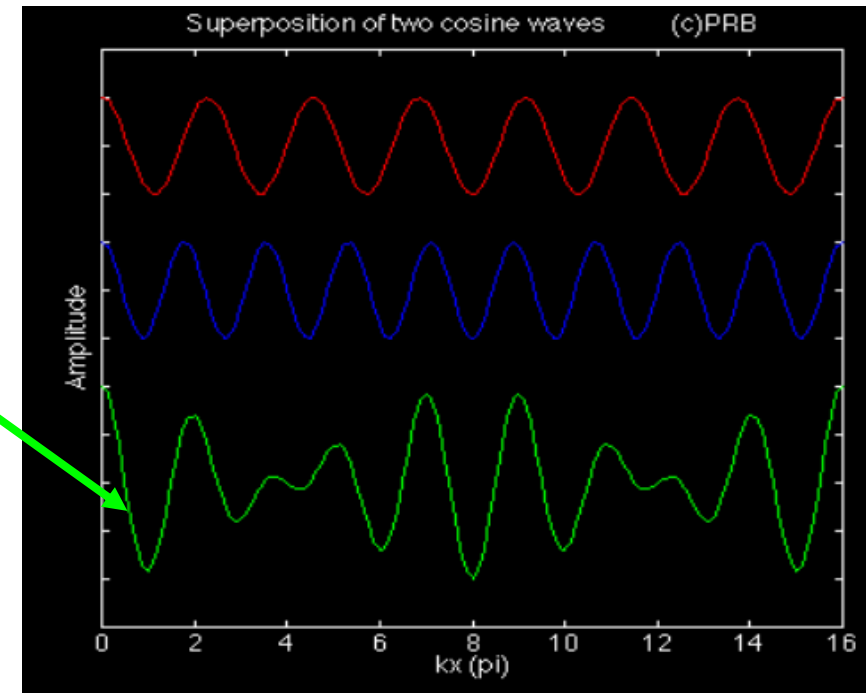
$$V_{Ph} = \frac{\omega}{k}$$

Group Velocity

Group Velocity (V_g): the velocity with which a wave packet (group of waves) travels is called **group velocity**.

- The relation between **group velocity** and **phase velocity** is **proportionate**.
- Schrodinger postulated that a moving particle (e-, etc) is equivalent to a wave packet rather than a single wave. Hence, **group velocity = particle velocity**.

$$V_g = \frac{d\omega}{dk}$$



Principle of Complementarity

In 1928, Niels Bohr told this principle of complementarity.

Principle: According to this all the objects in atomic dimension requires description of both wave and particle properties for complete knowledge about it, however particle and wave behavior are mutually exclusive. This dualism in quantum system is *principle of complementarity*.

The complementarity principle holds that objects have certain pairs of complementary properties which cannot all be observed or measured simultaneously.

Conclⁿ : But both wave and particle are not exhibited simultaneously.

Heisenberg's Uncertainty Principle



Heisenberg's Uncertainty Principle

Q: State and explain Heisenberg's Uncertainty principle. Show that the electron does not pre-exist inside the nucleus using uncertainty principle. (8M)

State:-“The position and momentum of a particle cannot be determined accurately and simultaneously. The product of uncertainty in the measurement of position and momentum is always greater than or equal to $\frac{\hbar}{2}$ (where, $\hbar = \frac{h}{2\pi}$)”.

$$\text{i.e.,} \quad \Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

Note : Similar to **Position - Momentum** uncertainty, there are other two uncertainty relations also present. They are: **Angular momentum - Angular displacement** uncertainty and **Energy - Time** uncertainty

$$\Delta L \cdot \Delta \theta \geq \frac{h}{4\pi}$$

$$\Delta E \cdot \Delta t \geq \frac{h}{4\pi}$$

Applⁿ of Heisenberg's Uncertainty Principle: **Non-existence of electron in the nucleus**

Background: This is related to the β -ray emission in nuclear physics. We know that the β -rays are actually streams of electrons, having kinetic energies in the range **3 to 4 MeV**.

Next question is “Is it possible that electrons exist inside the nucleus with certain energy.....???”. It can be answered by applying Uncertainty Principle as follows:

Proof: Energy of an electron of mass m moving with P can be expressed as,

$$\Rightarrow E = \frac{P^2}{2m} \quad (1)$$

from **Uncertainty Principle**, $\Delta p_x \cdot \Delta x \geq \frac{h}{4\pi} \Rightarrow \Delta p_x \geq \frac{h}{4\pi \Delta x}$ (2)

Wkt nucleus diameter is of the order 10^{-14} m. Thus, if electron is to exist inside the nucleus, then

$$\Delta x \leq 10^{-14} \text{ m} \quad \therefore \Delta p_x \geq \frac{6.626 \times 10^{-34}}{4 \times 3.142 \times 10^{-14}} \quad \text{or,} \quad \Delta p_x \geq 0.5 \times 10^{-20} \text{ N-s} \quad (3)$$

Now, by making use of this inequality (i.e., eqn. (3) in eqn. (1),

$$E \geq \frac{(0.5 \times 10^{-20})^2}{2(9.1 \times 10^{-31})} \Rightarrow E \geq 1.372 \times 10^{-11} \text{ J} \quad \text{or,} \quad E \geq \frac{1.372 \times 10^{-11}}{1.6 \times 10^{-19}} \text{ eV} \quad \boxed{E \geq 85 \text{ MeV}}$$

It indicates if electron exists inside the nucleus, then its energy $\geq 85 \text{ MeV}$. But, the β -particle KE is of the order of **3 to 4 MeV**. *This clearly indicates that, the emitted electrons cannot be the ones, which existed inside the nucleus*

Wave Function

A moving particle (electron, photon) with momentum p is described by a **matter wave**; its wavelength is

$$\lambda = h / p$$

A matter wave is described by a **wave function**: $\Psi(x, t)$

The general form of the total wave function is

$$\Psi = A e^{i(kx - \omega t)}$$

Separating the space and time dependent parts:

$$\Psi = A e^{ikx} e^{-i\omega t}$$

$\Rightarrow$

$$\Psi = \psi e^{-i\omega t} \quad | \quad \text{here } \psi = A e^{ikx} \text{ space dependent wave function}$$

In quantum mechanics, the **wave function** actually accounts for the wave like properties of a particle. It is obtained by solving fundamental eqn. called **Schrodinger Equation**.

Schrodinger Equation

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graph TD; A["Schrodinger Equation"] --> B["Time Dependent Schrodinger Equation"]; A --> C["Time Independent Schrodinger Equation"]; B --> D["Ψ(x, t)"]; C --> E["Ψ(x)"]
```

Time Dependent Schrodinger Equation

$$\Psi(x, t)$$

Time Independent Schrodinger Equation

$$\Psi(x)$$

Q: What is wave function? Arrive at the time independent Schrodinger wave equation. (8M)

Q: What is wave function? Arrive at the time independent Schrodinger wave equation. (8M)

According to de Broglie, $\lambda = \frac{h}{p} \Rightarrow P = \frac{h}{\lambda}$ (1)

The wave function describing this de Broglie wave,

$$\Psi = A e^{i(kx - \omega t)} \quad (2)$$

$$\Rightarrow \Psi = \psi e^{-i\omega t} \quad |\because \psi = A e^{ikx} \quad (3)$$

On diff **eqn. (3)** twice with respect to 'x', we get

$$\frac{d^2\Psi}{dx^2} = e^{-i\omega t} \frac{d^2\psi}{dx^2} \quad (4)$$

again diff **eqn. (3)** twice with respect to 't', we get,

$$\frac{d^2\Psi}{dt^2} = -\omega^2 e^{-i\omega t} \psi \quad (5)$$

wkt eqn. for travelling wave moving with vel 'v' & displ. 'y' as,

$$\frac{d^2y}{dx^2} = \frac{1}{v^2} \frac{d^2y}{dt^2} \Rightarrow \frac{d^2\Psi}{dx^2} = \frac{1}{v^2} \frac{d^2\Psi}{dt^2} \quad (6)$$

Substituting **eqns. (4) & (5)** in **eqn. (6)**, we get,

$$\cancel{e^{-i\omega t}} \frac{d^2\psi}{dx^2} = \frac{1}{v^2} (-\omega^2 \cancel{e^{-i\omega t}}) \psi \quad \text{or,} \quad \frac{d^2\psi}{dx^2} = \frac{-\omega^2}{v^2} \psi$$

But, wkt $\omega = 2\pi\nu$ and $v = \frac{\lambda}{T}$.

$$\therefore \frac{d^2\psi}{dx^2} = \frac{-4\pi^2\psi}{\lambda^2} \Rightarrow \frac{1}{\lambda^2} = -\frac{1}{4\pi^2\psi} \frac{d^2\psi}{dx^2} \quad (7)$$

For a moving particle, the **KE** is; $KE = \frac{1}{2}mv^2 = \frac{p^2}{2m}$

But, from **eqn. (1)**, wkt, $P = \frac{h}{\lambda}$ then above **eqn.** becomes,

$$KE = \frac{h^2}{2m} \frac{1}{\lambda^2} \quad (8)$$

Substitute the value of $\frac{1}{\lambda^2}$ from **eqn. (7)** in **(8)**, we get,

$$KE = \frac{-h^2}{8\pi^2m\psi} \frac{d^2\psi}{dx^2} \quad (9)$$

then total energy of the particle becomes, $E = KE + V$

$$\therefore E = \frac{-h^2}{8\pi^2m\psi} \frac{d^2\psi}{dx^2} + V \quad |\because \text{Using eqn. (9)}$$

$$\text{or,} \quad \frac{-h^2}{8\pi^2m\psi} \frac{d^2\psi}{dx^2} = E - V$$

$$\Rightarrow \frac{d^2\psi}{dx^2} + \frac{8\pi^2m}{h^2} (E - V)\psi = 0$$

This is the **time-independent Schrödinger equation.**

Physical Significance of Wave Function: Probability Density & Normalization

Probability density: If Ψ is the wave function associated with the particle, then $|\Psi|^2$ is the probability per unit volume or probability density that a particle will be found at the given region.

Max Born interpretation (Physical Significance): If τ be volume of certain region, inside which a particle is known to be present, but where exactly is not known (see **Fig.**). Then the probability of finding the particle in certain element of volume $d\tau$ of τ is always equal to " $|\Psi|^2 d\tau$ ".

Since, probability of occurrence of any event is a **real** and **positive** quantity. However, Ψ is **complex** & has no considerable physical significance. Hence, in order to get real and positive value,

$$|\Psi|^2 = \Psi \Psi^*$$

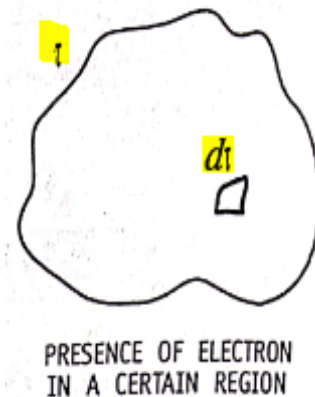
Normalization: If we are certain that, the particle is definitely present inside the volume τ then, as per the statistical rule,

$$\int_0^\tau |\Psi|^2 d\tau = 1$$

If we are not yet all certain about presence of particle in a finite volume then, the 'particle must exist somewhere in the space'.

$$\int_{-\infty}^{\infty} |\Psi|^2 d\tau = 1$$

This interpretation was first given by **Max Born** in 1926.



Properties of Wave Function

Q: Explain the physical significance of wave function. Mention its properties (6M)

- Ψ is **single valued** everywhere.
- Ψ is **finite** everywhere.
- Ψ and its **first derivative** with respect to its variable are **continuous**.
- For bound states, Ψ must **vanish at infinity** & hence $\Psi\Psi^*$ must vanish at infinity.

Eigen Function & Eigen Value

Def: “The wave functions that possess all the 4 properties of a wave function are named in quantum mechanics as **Eigen functions**”.

Note : **Operators in Quantum Mechanics:** It is a mathematical entity, which converts one function/state to other function/state.

The information of Ψ can be extracted by performing certain mathematical operations on Ψ .

Eigen Functions & Eigen values

Eigen values: When the operators are applied on **Eigen function** and produce Eigen function back along with a **constant** multiplied it then such constant values obtained for a physical observable are called as **Eigen values**.

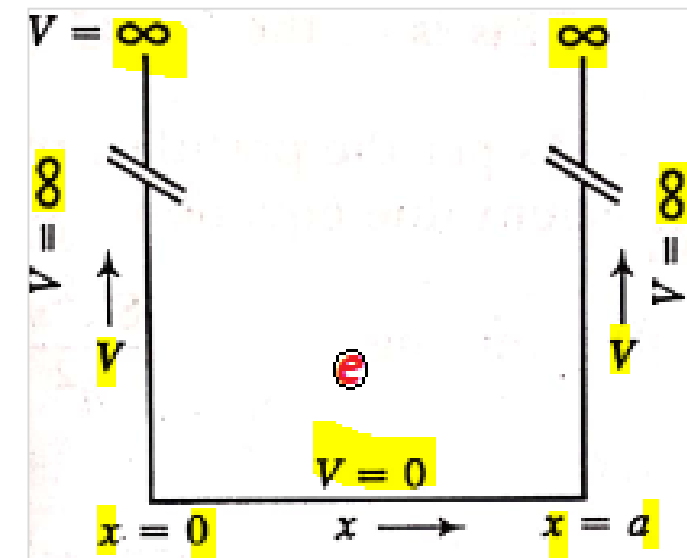
i.e., if an operator $\hat{A}$ acts on Eigen function Ψ and gives back Ψ along with Eigen value ' λ ', then

$$\hat{A}\Psi = \lambda \Psi$$

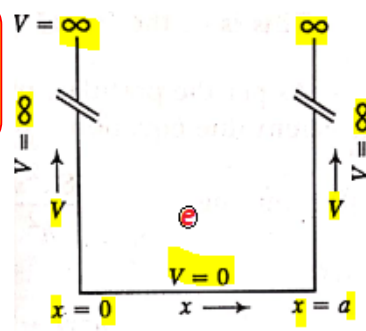
Eg:- (i) $\frac{d}{dx} (e^{ax}) = a e^{ax}$, (ii) $\frac{d^2}{dx^2} (\sin 2x) = -4 (\sin 2x)$, etc.

Application of Schrödinger wave equation: **Particle in 1-D potential well of infinite height:-**

Consider a particle of mass ' m ' is free to move along x-direction from $x = 0$ to $x = a$. In this region, its potential is $V = 0$. Outside this region, the potential is $V = \infty$. Such configuration of potential is called as **infinite potential well**.



Q: Using Schrodinger's time independent wave equation, obtain expression for energy Eigen value and normalized Eigen function for an electron present in 1D potential well of infinite height.(09 M)



Outside the well: the Schrödinger equation is given by,

$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2m}{h^2} (E - \infty)\psi = 0 \quad (1)$$

The above **eqn.** holds good only if $\psi = 0$ (I)

$\Rightarrow |\psi|^2 = 0$, indicates particle cannot be found at the wall and outside.

Inside the well: the Schrödinger equation is given by,

$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2m}{h^2} E\psi = 0 \quad (2)$$

In **eqn. (2)**, putting $\frac{8\pi^2m}{h^2} E = K^2$ (3)

We have, $\frac{d^2\psi}{dx^2} + K^2\psi = 0$

The solution of the above equation is given by,

$$\psi = C \cos Kx + D \sin Kx \quad (4)$$

But, $\psi = 0$ at $x = 0$ $\therefore$ I

$$\therefore \text{eqn. (4) becomes, } 0 = C \cos K \cdot 0 + D \sin K \cdot 0 \Rightarrow C = 0 \quad (5)$$

Again, $\psi = 0$ at $x = a$ $\therefore$ I

$$\therefore \text{eqn. (4) becomes, } 0 = C \cos Ka + D \sin Ka$$

$$\Rightarrow D \sin Ka = 0 \quad \therefore \text{using (5), } C=0$$

$$\text{Here, } D \neq 0, \therefore \sin Ka = 0$$

$$\Rightarrow Ka = n\pi \rightarrow K = \frac{n\pi}{a} \quad (6)$$

substituting the value of **eqn. (6)** in **eqn. (3)**, we get,

$$\frac{8\pi^2m}{h^2} E = \left(\frac{n\pi}{a}\right)^2 \Rightarrow E = \frac{n^2h^2}{8ma^2} \quad (8)$$

This is the required **Energy Eigen value**

Contd.....

And substituting the values of **eqn. (5)** and **(6)** in **eqn. (4)**, we get,

$$\psi = \textcolor{red}{D} \sin \frac{n\pi}{a} x \quad (7)$$

To find D: Consider the normalization of wave function as follows,

$$\int_0^a |\Psi|^2 d\tau = 1$$

Substituting Ψ_n from **eqn. (7)**, we have,

$$\int_0^a D^2 \sin^2 \left(\frac{n\pi}{a} \right) x dx = 1$$

$$\Rightarrow D^2 \left[\int_0^a \frac{1}{2} \left[1 - \cos 2 \left(\frac{n\pi}{a} \right) x \right] dx \right] = 1$$

$$\text{i.e., } \frac{D^2}{2} \left[\int_0^a dx - \int_0^a \cos \left(\frac{2n\pi}{a} \right) x dx \right] = 1$$

$$\text{or, } \frac{D^2}{2} \left[[x]_0^a - \frac{a}{2n\pi} \sin \frac{2n\pi x}{a} \Big|_0^a \right] = 1$$

$$\frac{D^2}{2} \left[a - \frac{a}{2n\pi} \sin 2n\pi \right] = 1$$

$$\text{Or, } \frac{D^2}{2} [a] = 1 \quad |\because \sin 2n\pi = 0$$

$$\therefore \textcolor{red}{D} = \sqrt{\frac{2}{a}}$$

Putting the value of ' D ' in **eqn. (7)**, we get,

$$\psi_n = \sqrt{\frac{2}{a}} \sin \left(\frac{n\pi}{a} \right) x$$

This is the **Eigen function** for a particle in an infinite potential well.

Eigen functions, Probability densities and energy Eigen values for a particle in Potential well

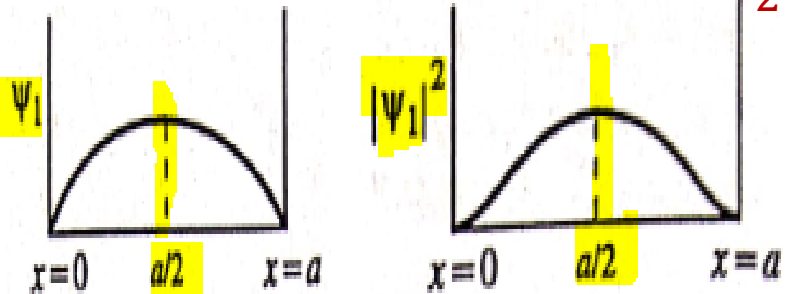
Case (i): $n = 1$:- ground state

Eigen function becomes;

$$\psi_1 = \sqrt{\frac{2}{a}} \sin\left(\frac{\pi}{a}\right) x$$

$\psi_1 = 0$ at $x = 0$ and $x = a$

But, $\psi_1 = \text{max}$ at $x = \frac{a}{2}$



ground state energy for $n=1$;

$$E_1 = \frac{h^2}{8ma^2} = E_0$$

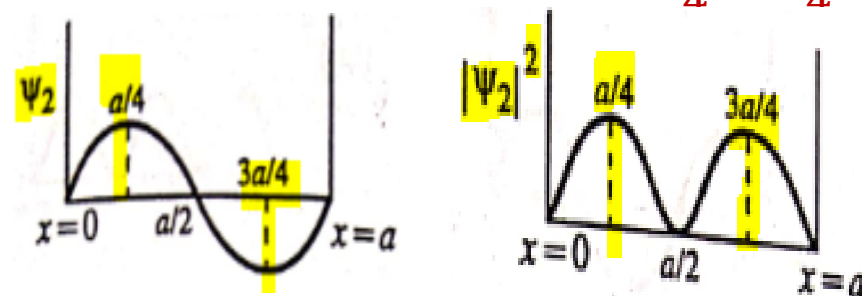
Case (ii): $n = 2$:- first excited state

Eigen function becomes;

$$\psi_2 = \sqrt{\frac{2}{a}} \sin\left(\frac{2\pi}{a}\right) x$$

$\psi_2 = 0$ @ $x = 0, \frac{a}{2}$ and a

But, $\psi_2 = \text{max}$ @ $x = \frac{a}{4}$ & $\frac{3a}{4}$



First excited state energy for $n=2$;

$$E_2 = \frac{4h^2}{8ma^2} = 4E_0$$

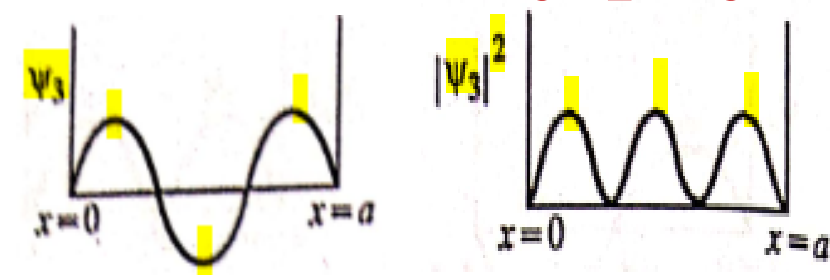
Case(iii): $n = 3$: second excited state

Eigen function becomes;

$$\psi_3 = \sqrt{\frac{2}{a}} \sin\left(\frac{3\pi}{a}\right) x$$

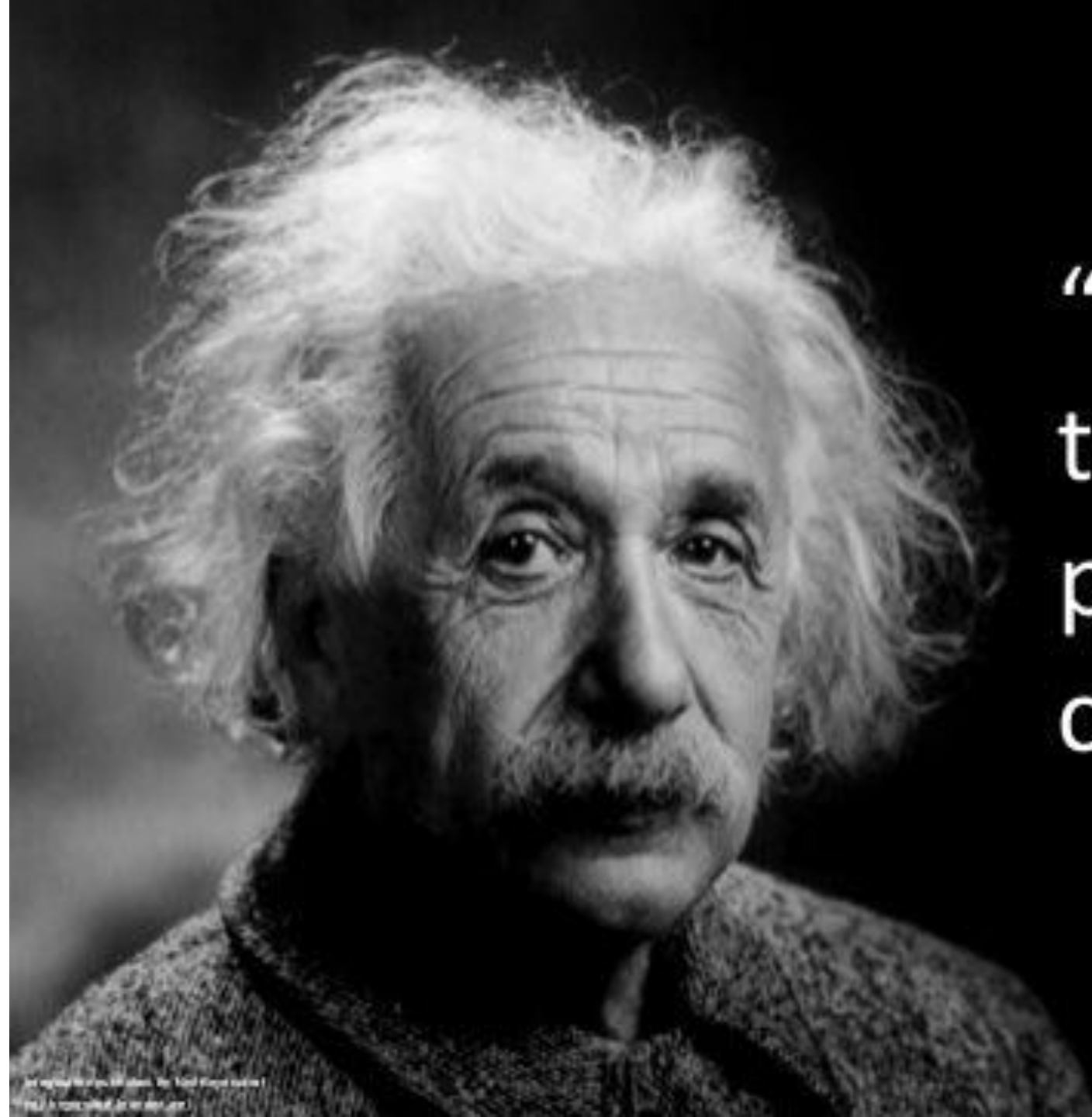
$\psi_3 = 0$ @ $x = 0, \frac{a}{3}, \frac{2a}{3}$ & a

$\psi_3 = \text{max}$ @ $x = \frac{a}{6}, \frac{a}{2}$ & $\frac{5a}{6}$



Second excited state energy for $n=3$

$$E_3 = \frac{9h^2}{8ma^2} = 9E_0$$



“I have no special talents. I am only passionately curious.”

Albert Einstein